

UNION COUNTY AP, OH, USA

WMO: 720651

Lat: 40.225N

Lon: 83.352W

Elev: 311

StdP: 97.64

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00232

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-14.6	-22.0	0.5	-17.7	-18.4	0.8	-13.5	12.3	-1.9	11.1	-1.6	2.8	270	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.2	22.5	30.9	21.9	29.7	21.4	24.4	29.1	23.6	28.3	22.9	27.4	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	26.8	22.2	17.6	26.3	21.4	16.7	25.4	75.0	29.2	72.2	28.4	69.1	27.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.4	8.7	7.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-22.4	34.0	4.6	1.7	-25.7	35.2	-28.4	36.2	-30.9	37.2	-34.3	38.5
			-22.7	25.7	4.6	0.9	-25.9	26.4	-28.6	26.9	-31.2	27.5	-34.5	28.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-3.3	-1.3	4.0	10.8	17.6	21.5	23.1	21.5	18.4	12.0	5.1	0.7	(6)
(7)		DBStd	10.45	6.77	6.92	6.01	4.91	4.92	3.15	2.88	2.71	4.19	5.04	5.12	5.45	(7)
(8)		HDD10.0	1481	414	321	203	48	4	0	0	0	1	36	162	292	(8)
(9)		HDD18.3	3189	671	550	444	228	75	10	2	6	51	206	398	548	(9)
(10)		CDD10.0	1807	1	4	18	72	239	344	406	355	254	97	14	3	(10)
(11)		CDD18.3	476	0	0	0	3	52	104	149	103	54	9	0	0	(11)
(12)		CDH23.3	4113	0	0	4	50	468	855	1285	834	516	98	2	0	(12)
(13)		CDH26.7	1269	0	0	1	4	127	269	456	212	176	23	0	0	(13)
(14)	Wind	WSAvg	3.2	4.0	3.8	3.8	3.8	3.2	2.9	2.2	2.0	2.2	3.2	3.4	3.5	(14)
(15)	Precipitation	PrecAvg	1020	77	55	73	93	117	114	113	89	75	67	72	76	(15)
(16)		PrecMax	1288	239	112	159	176	202	215	240	169	156	138	131	208	(16)
(17)		PrecMin	692	30	21	24	29	50	42	46	23	19	19	21	24	(17)
(18)		PrecStd	146	45	22	34	35	47	47	45	42	41	32	31	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	19.1	23.6	27.3	31.4	32.8	33.9	31.9	33.0	29.8	21.5	16.6	(19)
(20)			MCWB	12.9	14.0	16.2	16.4	20.9	22.0	23.7	22.5	21.7	20.5	15.4	14.4	(20)
(21)		2%	DB	12.5	14.8	19.7	24.8	29.8	31.3	32.3	30.3	30.9	26.0	18.4	13.8	(21)
(22)			MCWB	10.7	12.3	14.3	15.8	19.8	21.6	23.4	22.3	21.6	17.9	13.0	11.9	(22)
(23)		5%	DB	8.9	12.1	16.4	22.2	27.9	29.8	30.9	29.0	28.8	23.4	15.9	11.7	(23)
(24)			MCWB	7.8	10.3	11.8	14.4	19.3	20.6	22.5	21.9	20.8	16.7	11.1	10.3	(24)
(25)		10%	DB	6.1	8.0	13.8	19.5	26.1	27.9	29.3	27.7	26.4	20.8	13.3	8.8	(25)
(26)			MCWB	4.5	5.8	10.6	13.7	18.4	20.1	22.0	21.1	19.0	15.9	9.8	7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	15.3	16.8	18.7	22.4	24.2	25.7	25.1	23.8	21.6	16.7	14.9	(27)
(28)			MCDWB	15.1	18.5	22.7	25.0	28.2	29.7	31.2	29.1	29.5	27.1	20.7	16.3	(28)
(29)		2%	WB	10.7	12.7	15.2	16.8	21.4	23.4	24.8	24.0	22.8	19.9	14.3	12.2	(29)
(30)			MCDWB	12.0	13.8	19.0	21.7	27.4	28.7	29.9	28.2	28.8	23.6	16.8	13.4	(30)
(31)		5%	WB	7.9	9.8	13.0	15.7	20.5	22.4	23.9	23.2	21.8	18.3	12.0	10.3	(31)
(32)			MCDWB	8.7	12.5	15.1	20.2	26.1	27.3	28.5	27.0	26.8	21.6	14.8	11.6	(32)
(33)		10%	WB	4.6	6.1	11.0	14.5	19.5	21.5	23.1	22.3	20.7	16.3	10.0	7.2	(33)
(34)			MCDWB	5.9	8.0	13.4	18.7	24.4	26.0	27.6	25.9	24.3	20.7	13.1	8.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	9.9	12.1	11.4	10.8	10.7	11.6	12.6	11.8	10.1	7.4	(35)
(36)			MCDBR	10.4	12.3	13.9	15.5	13.4	13.1	12.2	12.8	14.9	15.3	14.2	10.3	(36)
(37)			MCWBR	9.4	9.9	8.7	8.3	5.5	5.3	4.7	5.9	6.4	7.8	9.2	9.2	(37)
(38)		5% WB	MCDBR	10.4	11.9	11.7	12.7	11.2	10.8	10.6	10.1	12.2	12.1	11.6	9.8	(38)
(39)			MCWBR	9.8	10.1	8.3	8.3	5.2	5.4	5.1	5.2	5.7	6.8	8.7	9.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.326	0.355	0.395	0.435	0.448	0.454	0.449	0.391	0.359	0.326	0.301	(40)	
(41)		taud	2.410	2.341	2.333	2.244	2.173	2.181	2.176	2.173	2.316	2.392	2.446	2.454	(41)	
(42)		Ebn at Noon	855	886	896	878	847	834	824	818	846	837	825	834	(42)	
(43)		Edh at Noon	82	101	114	134	147	146	146	142	115	96	79	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.63	2.52	3.49	4.71	5.50	6.04	5.95	5.43	4.41	3.03	2.03	1.38	(44)	
(45)		RadStd	0.15	0.30	0.32	0.40	0.45	0.43	0.35	0.25	0.38	0.27	0.26	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+68

Nomenclature: See separate page